

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM

451/1

COMPUTER STUDIES

PAPER 1 (THEORY)

TIME: 2½ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

Instructions to Candidates

- (a) Write your name and index number in the spaces provided above.
- (b) This Paper consists of **two** sections **A** and **B**.
- (c) Answer **all** the questions in **section A**.
- (d) Answer **question 16** and any other **three** questions from **section B**.
- (e) **All** answers should be written in the spaces provided.

FOR EXAMINER'S USE ONLY

Section	Questions	Candidate's score
A	1-15	
B	16	
	TOTAL SCORE	

SECTION A: (40 MARKS)

Answer ALL questions in this section

1. Name any **two** computing devices developed during pre-electronic age. **(2marks)**

.....
.....

2. Several items of data are input to a computer using direct data entry methods. Write down the most appropriate input device to input each item of data.

a) Data written in special ink at the bottom of bank cheques. **(1mark)**

.....

b) Information on the label of a supermarket product. **(1mark)**

.....

c) Shaded in boxes on a student's examination answer sheet. **(1mark)**

.....

d) Debit card details input at an ATM. **(1mark)**

.....

3. Differentiate between a monitor and a printer in terms of information output. **(2marks)**

.....
.....
.....
.....

4. Kilimambogo has decided to create a new file using an application program. He added content in the file. Identify the type of storage device that he was using while creating the file. **(1mark)**

.....

5. Kilimambogo decided to save the file after adding content. Identify the type of storage device he used. **(1mark)**

.....

6. Describe **three** types of computer buses (3marks)

.....
.....
.....
.....
.....

7. Describe the following concepts as used in applications.

i) Automatic recalculation. (2marks)

.....
.....
.....

ii) What if analysis. (2marks)

.....
.....
.....

8. Password is a secret code for protecting systems against unauthorized access. State **three** features of a password.

(3marks)

.....
.....
.....

9. Explain the following terminologies used in an e-mail software.

(i) CC (2marks)

.....
.....
(ii) BCC **(2marks)**
.....
.....

10. Identify the data processing modes that are used in performing the following activities:

(i) Processing examination marks for students in a school. **(1mark)**
.....

(ii) Booking a bus ticket using computing device and choosing a specific seat. **(1mark)**
.....

(iii) Buying of data bundles using a software that provides the user with immediate response to enable the user to decide and make a selection for the next input. **(1mark)**
.....

11. Name **three** types of non-printing guides in DTP.
(3marks)
.....
.....
.....

12. State **two** advantages of USB ports over COM ports. **(2marks)**
.....
.....

13. A supermarket decided to computerize their system. State **three** benefits that the supermarket would realize by doing this. **(3marks)**
.....
.....

.....

14. Kimani wanted to buy an application software, he was advised to buy an integrated package. What is meant by the term integrated software. (2marks)

.....
.....
.....

15. Explain **three** benefits that Kimani would experience by using an integrated software other than the single package. (3marks)

.....
.....
.....
.....
.....
.....
.....

SECTION B: (60 MARKS)

Answer Question 16 and any other three questions from this section

16. (a) Explain the term structured programming as used in program development. (2marks)
- (b) Name the stage of program development in which: (4marks)
- (i). A flowchart would be drawn.
 - (ii). The programmer would check whether the program does as required.
 - (iii). User guide would be written.
 - (iv). The requirement specification would be written.
- (c) State **two** function of an assembler. (2marks)
- (d) To qualify to play in football tournament, a student must be aged between 13 and 18, the system automatically checks a player’s age once their names and date of birth have been provided. 11

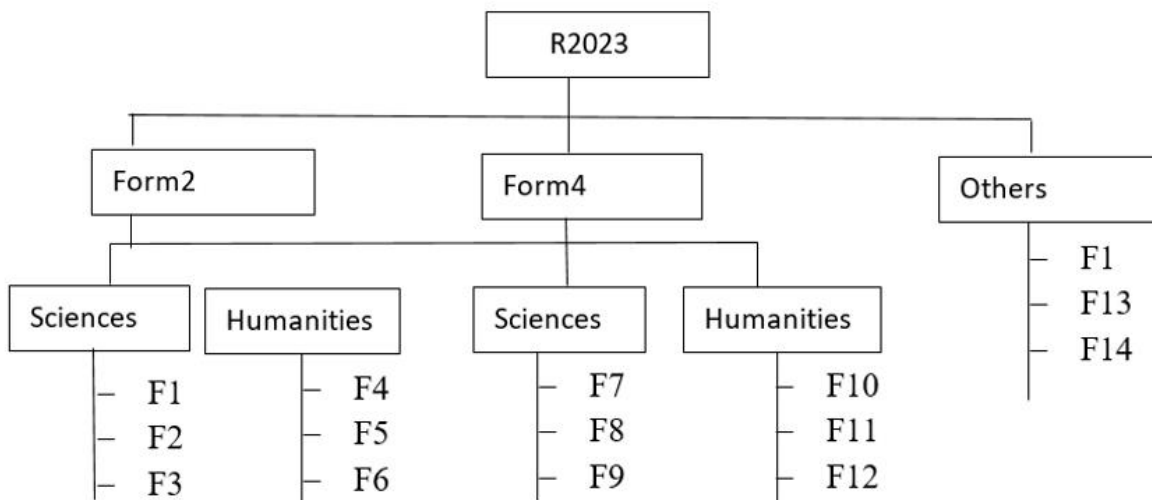
participants were evaluated and their qualification status displayed alongside their name and age.

Draw a flowchart that would be used to create the program. (7 marks)

17. a) . State **three** ways in which an operating system performs error handling in a computer system.

(3marks)

b). An operating system organizes files in directories as shown in the chart below. Study it and answer the questions that follow.



(i) Write the name of this file structure? (1mark)

(ii) Explain what would happen if an attempt is made to delete **Form4** while **F10** in **Humanities** is opened. (2marks)

(iii) Write the path of the **F6**. (2marks)

(c) Describe **three** activities that are performed in the maintenance phase during system development.

(6 marks)

(d) State **one** role of an information system analyst. (1mark)

18. (a) What is meant by the term data communication? (1mark)

(b) Describe **three** data communication elements (3marks)

(c) State the meanings of the following network and internet terms: (5marks)

- i. Protocol
- ii. Client
- iii. Server
- iv. Home page
- v. DTE

(c) With aid of a diagram, describe the star network topology **(6marks)**

19 (a) Define each of the following computer terminologies **(2marks)**

a. Nibble

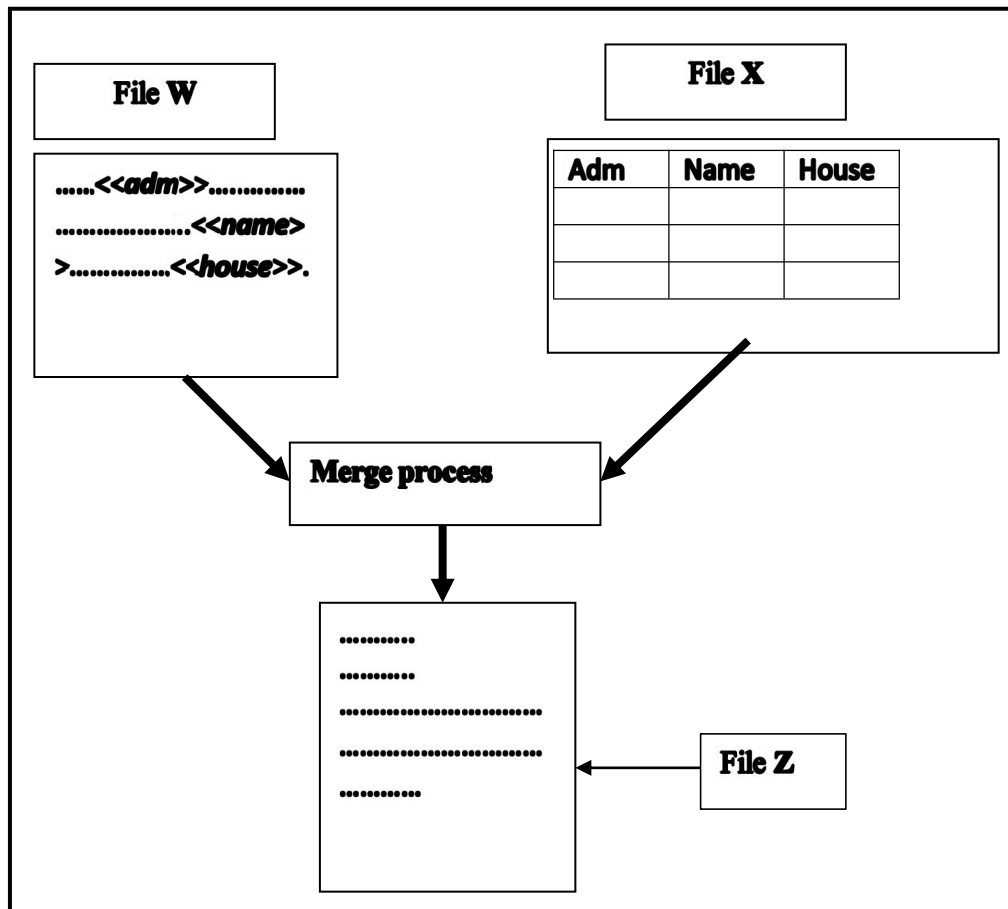
b. Byte

(b) State **three** ways of reducing Repetitive Strain Injuries when using computers. **(3marks)**

(c) Using twos compliment, perform the following binary arithmetic leaving your answer in decimal notation.

$$1101_2 - 100101_2 \quad \textbf{(6 marks)}$$

(d) The diagram below shows how mail merging is done in word processing. Use it to answer the questions below.



Identify the files labeled:

(3marks)

(i) **W**

(ii) **X**

(iii) **Z**

(e) Name the terminology used to refer to items enclosed in << >> symbols in file **W**. (1mark)

20. (a) Explain **three** components of an Expert System.

(6marks)

(b) The following is an extract from a database obtained from a National organization. Use the table to answer questions which follow:

S/NO	EMPLOYMENT ID	NAME	AGE	DEPARTMENT	SALARY(SH)	STATUS
001	EMP01	Lucy	40	Secretary	40,000	Married
002	EMP02	Agnes	45	Cook	20,000	Single
003	EMP03	Peter	30	Messenger	15,000	Married

004	EMP04	Joel	35	Account	50,000	Married
005	EMP05	Abdi	20	Sales	45,000	Single
006	EMP06	Baraka	25	Security	10,000	Single

(i.) Which field could best fit to be a primary key?

(1mark)

(ii) Explain your answer in (i).

(2marks)

(iii) Describe how they would extract all the details of those who are 30 years and above and are married.

(3 marks)

(c) State **three** area where virtual reality can be applied.

(3marks)